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Transformer-aided skin cancer classification using VGG19-based feature encoding

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Skin cancer is among the most widely distributed, deadliest cancers around the globe, and early diagnosis becomes vital to enhance patient survival. Deep learning has demonstrated high potential for automatic skin lesion classification. However, existing Convolutional Neural Networks (CNNs) are still unsatisfactory in terms of dataset reliance, subjection to orientations, and the inability to model long-range global context. To solve this problem, we propose a hybrid model named the VGG19-RSPDA-ViT with the fine-grained local feature captured by VGG19 and the global context provided by Vision Transformers (ViT). The proposed RSPDA enforces rotation invariance and enriches the feature space, which further strengthens generalization on a small training set. To the best of our knowledge, this is the first work to systematically combine feature-map-level rotation/shift augmentation with a CNN-Transformer hybrid model for dermoscopic skin cancer detection. Performance was validated on three benchmark datasets: the Melanoma Skin Cancer Dataset of 10,000 Images (MSK10000) for binary classification and the Human Against Machine with 10,000 training images (HAM10000) and Hospital Pedro Hispano (PH2) for multi-class classification. Our model achieved accuracies of 97.9%, 97.1%, and 98.67% on MSK10000, HAM10000, and PH2 datasets, respectively, with consistently high macro-averaged precision, recall, specificity, and F1 scores across both datasets. VGG19-RSPDA-ViT outperformed existing state-of-the-art methods with superior generalization capabilities. These results demonstrate that our proposed model is effective for skin lesion classification and has significant potential for clinical application as an automated diagnostic tool in dermatology.

Keywords Skin cancer, VGG19-RSPDA-ViT, Image augmentation, Skin lesion classification

Skin cancer is a major global health challenge and is one of the most common cancers worldwide[1]. The World Health Organization (WHO) reports that millions are diagnosed every year, with melanoma responsible for a significant number of skin cancer deaths[2]. Melanoma is an especially aggressive type of skin cancer that can quickly spread to other parts of the patient's body if it is not detected early[3]. Diagnosis of this syndrome depends on dermoscopic imaging, but a large variation in the shape, color and texture of the lesion makes expert evaluation difficult[4,5]. Deep learning techniques, notably Convolutional Neural Networks (CNNs) like VGG, ResNet, and Inception, achieve the state-of-the-art performance on benchmark datasets like the International Skin Imaging Collaboration (ISIC), Melanoma Skin Cancer Dataset of 10,000 Images (MSK10000), the Human Against Machine with 10,000 training images (HAM10000), and Hospital Pedro Hispano (PH2). These models perform well for local texture and edge patterns, reaching classification accuracy that is nearly comparable to expert dermatologists[6]. Recently, a Vision Transformer (ViT) has been proposed for medical image analysis, including dermoscopy and other skin imaging tasks, utilizing a global self-attention mechanism for learning the long dependencies among patches. There has also been promising work on hybrid CNN-Transformer models (e.g., TransMed, Swin-UNet), mostly in the segmentation and multimodal settings.

Nevertheless, Dermoscopic images, providing a detailed view of the skin surface micropatterns, are often used to diagnose skin lesions[7]. Furthermore, dermoscopic analysis remains challenging because skin lesions

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exhibit considerable morphological variation[8,9], such as shape, size, color, and texture. For example, nevi (moles) belong to benign lesions but can resemble early-stage melanomas, complicating the interpretability[10]. Other variables include skin tone, lighting conditions, and the anatomical location of the lesion. These variations necessitate more accurate and automated skin lesion classification systems to aid dermatologists in decision-making. Recently, deep learning has become a state-of-the-art method to automate skin cancer diagnosis with accuracy comparable to that of trained dermatologists in image classification tasks[11].

CNNs make an impressive contribution to medical imaging based on their hierarchical level depth of input image learning, allowing deep representation of the learning approach[12,13]. CNNs can learn how to recognize each type of skin lesion automatically from low-level features, such as local edges, textures, and patterns[14–17]. Models like ResNet, Inception and VGGs have been widely applied in skin lesion classification and achieved state-of-the-art results on benchmark datasets, such as the International Skin Imaging Collaboration (ISIC) archive[18]. One of the most adopted networks for feature extraction is VGG19, proposed by the Visual Geometry Group at the University of Oxford, which is simple and powerful[19]. VGG19 has 19 layers, of which 16 are convolutional, and three are fully connected, making it ideal for extracting fine-grained features from dermoscopic images. Its consistent design with a small 3×3 convolutional filter size enables deeper levels of feature learning, which is useful in image characteristics measurement[20].

Nevertheless, although their filters can extract important visual structures and fine-grained local features, VGG19 and other CNNs fundamentally lack global context due to their localized convolution filters. Adding more layers in CNNs can partially resolve this constraint by extending the receptive field[21]. However, this approach also risks overfitting and increased computational cost, particularly when the datasets are small or highly variable[22].

In order to overcome the limitations of CNNs, ViT has been proposed as a new paradigm for image classification[23]. Initially motivated by the advances of transformers in Natural Language Processing (NLP), ViT demonstrated that self-attention mechanisms can be successfully transferred to the image analysis domain[24]. In contrast to CNNs, ViT passes a convolutional kernel over the input image slidingly and treats an image as a sequence of patches. Fixed-size patches are used to partition each image, which is then flattened and linearly embedded into tokens for the transformer to process using multi-head self-attention[25]. ViT's global attention mechanism enables it to learn long-range dependencies between patches, capturing information on the entire image context, which is often tougher for CNNs to learn. For diverse medical images, in which the spatial relationships between regions are critical, ViT can effectively master intricate relationships among regions throughout the whole image. Despite its advantage, applying ViT directly to medical image analysis remains a challenge[26].

One of the major limitations is their large data requirement: ViT often requires large-scale datasets to learn better representations. Medical imaging datasets, such as skin lesions, are generally small since acquiring and annotating high-quality medical images is meticulous[27]. Moreover, ViT can be more sensitive to image orientation and scale since they do not have the inductive bias that makes CNNs robust to such variances[28]. This sensitivity can be especially problematic when dealing with skin lesions, which can be inspected in multiple orientations, i.e., the same lesion can be imaged in many ways when looking at either arm or the scale. Moreover, delicate skin image details might get lost due to differences in distance or angle used to capture the image. Consequently, the performance of a ViT-based method is highly contingent on the augmentation and preprocessing methods employed[29].

Data augmentation is a common technique for enhancing model generalization by introducing variation in training data. Augmentations, such as rotations, flips, color adjustments, and translations, have been used in skin lesion classification to make the model more robust to variations[30]. On the other hand, simple augmentation fails to derive spatial correlation between different lesion areas. For instance, a melanoma lesion may have wavering edges and a color variegation pattern that must be scanned in multiple directions to ascertain that the model captures the subtle malignancy features[31]. Flexibility in dimensions becomes an obstacle in the localization task as the spatial aspect is still a key point, and the scarcity of data presents a difficulty for both CNNs and transformers.

To mitigate these problems, we have developed a compound model for skin lesion classification, denoted as VGG19-RSPDA-ViT. First, a pretrained VGG19 is used as a backbone to capture fine-grained dermoscopic features. Then, our new RSPDA can produce orientation-invariant feature-map views in the second-hand stage to increase spatial diversity. These augmented features are tokenized and fed into a light ViT encoder that learns contextual relationships among lesion areas. Finally, a fully connected classifier heads that predict lesion category in both binary and multi-class. This model combines the strengths of CNNs and Transformers while addressing their weaknesses.

Our main contributions are: (i) introducing Rotated and Shifted Patch Data Augmentation Vision Transformer (VGG19-RSPDA-ViT), a novel hybrid model that combines VGG19 and ViT to enhance the accuracy of skin lesion classification. (ii) Integrate the well-established VGG19 architecture to extract rich local features that capture the fine-grained textural details of skin lesions. (iii) Incorporate the RSPDA tokenization strategy within the ViT, which enables the processing of augmented feature maps through rotation and shift operations that enhance training data diversity and introduce rotational invariance. Finally, significant improvements in skin lesion classification accuracy were demonstrated through: (iv) Minimizing overfitting while maintaining consistency across different benchmark datasets, and (v) effectively addressing orientation sensitivity and the limited availability of annotated data.

Although few hybrid CNN-Transformer architectures have been applied in medical imaging (e.g., TransMed, MedT, Swin-UNet), they are mostly focused on segmentation tasks or multimodal analysis, which involves working with more than one type of information and is dependent on complex hierarchical transformer structures. Thus, our work focuses on dermoscopic skin lesion classification, addressing two main challenges: (i)

fine-grained local feature extraction using VGG19 for its robust performance in dermoscopy, and (ii) orientation ambiguity of skin lesions through integrating RSPDA-augmented features into ViT to ensure rotational invariance and improve performance gains with fewer data samples. Additionally, while other recent works tend to overfit to a single dataset, we show effectiveness on three datasets: the MSK10000, binary classification, and the HAM10000 and PH2, multi-class classification, with competitive cross-dataset performance.

Related works

In the evolving field of skin cancer diagnosis, researchers have developed various methods, including deep learning and machine learning approaches, to improve the accuracy and speed of melanoma detection. The innovation process begins with the work of Subhayu Ghosh et al.[32], who proposed an ensemble model to combine the advantages of DCNNs, Capsule Networks (Caps-Net), and ViT for the process of feature extraction. These extracted features were used as input to classical classifiers (Random Forest, SVM, XGBoost), and an ensemble model, achieving 91.6% accuracy on the MSK10000 dataset. However, while the model relies on handcrafted ensemble mechanisms that are difficult to interpret, this dependency may limit real-time adaptability in clinical applications. Focusing on multimodal learning, Adebiyi et al.[33] filled the gap for such a context by introducing dermoscopic images and patient metadata as a multimodal input to a deep learning model. Tested on HAM10000 and ISIC2017 datasets, their model achieved superior performance compared to unimodal models with an accuracy of 94.11% and an AUC of 0.9426 on HAM10000. Although this method offers valuable real-life applicability, incomplete clinical information may limit its effectiveness in resource-constrained settings.

Furthermore, Ingle and Shaikh compared the performance of plain CNNs with VGG16 on HAM10000 data. The authors found that VGG16 performed better with the use of data augmentation and regularization, achieving an accuracy of 88.83%. This study revealed the strength of transfer learning and increased network depth for multi-class categorization problems, although class imbalance and inter-class overlap continue to pose difficulties[34]. Similarly, Wicaksana et al. investigated the single-stage detection model YOLOv11 with the HAM10000 dataset. YOLOv11x-cls attained 84.74% accuracy, while VGG19 achieved 89.68%, demonstrating that multi-stage architectures are superior in fine-grained classification. Their results also highlighted the effect of data augmentation in addressing class imbalance and revealed the limitations of single-stage networks[35].

In contrast, Sinchana et al.[36] proposed a hybrid CNN model with ResNet and VGG backbones using transfer learning on the MSK10000 training dataset. They investigated the choice of optimizer (RMSProp, Adam or SGD) and the performance measure (specificity, sensitivity). Although the model showed robust performance, its generalizability remained uncertain, requiring validation on external cohort datasets. Furthermore, Sabir and Mehmood[37] conducted a comparative study of CNN, ResNet-18, and EfficientNet-B0. EfficientNet-B0 appeared to be the most promising and performed with an accuracy of 97%, a precision of 95%, and a sensitivity of 99%. This work confirmed the potential of the model and also observed its sensitivity to hyperparameter settings and quality of preprocessing with rigorous metric analysis. Al Huda et al.[38] developed DeepMelaNet, designed for early-stage melanoma detection. This model, trained on the MSK10000 dataset, achieved 93.40% accuracy. Their novel deep-learning architecture achieved balanced F1-scores across all classes. However, the study also reported several difficulties in stage-specific categorization and dataset diversity.

Bhuvaneswari and Raman evaluated AlexNet, and it achieved only 91.1% compared with InceptionV3 and VGG19. Although this work demonstrated the continuing utility of AlexNet in medical imaging, which represents lesion complexity due to a small field of view and retained features, its relatively shallow depth and hard feature extraction hamper its ability to address complex lesion patterns[39]. Srivastava et al.[40] made significant advances in ensemble learning through VCCINet, a multi-model ensemble combining CNN, CapsNet, InceptionV3 and feature-extracted VGG16. Their refined model, FT-VCCINet, integrated pre-trained knowledge and achieved better performance than the single model. The ensemble model delivered consistent but marginal gains across precision, recall, F1-score, and specificity, although the performance fluctuations across training epochs revealed the model's sensitivity to training schedules.

To balance local texture encoding and global context modeling in dermoscopic image analysis, recent studies have progressively integrated convolutional and transformer paradigms. Aruk et al.[41] combined a ConvNeXt-CAFormer hybrid, whereby ConvNeXt blocks extract fine-grained local patterns and transformer stages capture long-range dependencies. Under identical training conditions to strong CNN and ViT baselines on HAM10000 and ISIC 2019, the authors report that this hybrid achieves 94.30% accuracy and 91.11% F1 on HAM10000, and 92.50% accuracy and 90.38% F1 on ISIC 2019, outperforming more than twenty competitors.

In contrast, Yang et al.[42] proposed the ViTfSCD pipeline, a ViT-based architecture customized for dermoscopy comprising four phases: data augmentation and class rebalancing, patch tokenization, transformer encoder, and batch normalization regularized classifier. They attained 94.1% accuracy with HAM10000, demonstrating that oversampling/undersampling combined with duplicate removal are critical. They also highlighted ViT's data-efficiency limitations, which often demand large-scale pre-training and illustrate open issues in robustness as well as generalization beyond curated dermoscopy. All together, these results establish class rebalancing and duplication control among the first-order considerations for deploying a transformer in dermatology.

Besides CNN-based methods, several hybrid CNN-Transformer architectural designs have been recently introduced in the field of medical imaging. There had already been some successful examples demonstrating the advantages of integrating CNNs with Transformers for multimodal learning (e.g., TransMed, MedT) and medical image segmentation tasks. At the same time, hierarchical self-attention has also successfully been applied to medical images through Swin-UNet variants. While useful in their own right, these techniques are largely targeted either for segmentation or multimodal tasks and rely on deeper hierarchical architectures. In contrast, our skin lesion classification for dermoscopy does not require ImageNet pre-training, uses a low-resource and interpretable design (incorporating VGG19 feature extraction into the proposed lightweight Vision Transformer powered with RSPDA), demonstrating promising results on both robustness and generalization.

Lastly, Waheed et al.[7] introduced a CNN-based architecture designed for melanoma diagnosing. They attempted to solve the problem by preprocessing and augmenting the noisy and various image inputs. The approach yielded high diagnosis accuracy and was promising for low-resource implementation. However, its internal validation introduces the potential for overfitting and external generalizability.

Further, previous works have shown significant results in skin lesion classification, but they still have limitations. Ensemble-based methods, such as those by Ghosh et al.[32], achieve high accuracy but face challenges in interpretability and increased computational burden, making them unsuitable for clinical deployment. Multimodal approaches by Adebiyi et al.[33] demonstrated the importance of clinical metadata. Nevertheless, the requirement for complete clinical data, coupled with patient privacy considerations, may limit practical applicability in resource-constrained settings[34].

Wicaksana et al.[35] showed that deeper architectures like InceptionV3 and GoogLeNet achieve superior performance by capturing fine-grained lesion texture patterns; these models remain biased due to class imbalance and have difficulty generalizing to underrepresented lesion types. Regarding transfer learning approaches, studies by Sinchana et al.[36] and Sabir and Mehmood[37] showed high recall, but their generalizability on external validation datasets remains unknown.

Recent works have presented some results on PH2. For example, Ghazouani[43] proposed the Multi-Residual Attention Network (MuRANet), which combines multi-scale dilated, spatial, channel-wise and squeeze-and-excitation attention strategies. Their method attained 97–98% accuracy and F1-score on PH2, which was considered one of the best. In contrast, Golkarieh et al.[44] achieved lower accuracy with 89.53% and a recall of 90.97%, as well as an F1-score of 90.62%, which suggests that it is not feasible to incorporate lesion variability properly.

In comparison to these works, our VGG19-RSPDA-ViT obtained an accuracy of 98.67% and a F1-score of 97.72% on PH2, which is slightly superior to MuRANet. These comparisons demonstrate the competitiveness of attention-augmented deep learning strategies on PH2 and the effectiveness of our hybrid CNN-Transformer framework for modelling local and global lesion features. Finally, our developed VGG19-RSPDA-ViT method addresses these limitations by combining fine-grained CNN feature extraction with transformer-based global context modeling. The RSPDA augmentation strategy provides rotational invariance while improving classification performance on imbalanced datasets.

Methodology

The proposed VGG19-RSPDA-ViT model combines the strengths of CNNs and ViT to improve skin lesion classification. The methodology comprises three primary stages: feature extraction using the VGG19 backbone, rotated and shifted patch tokenization, and ViT encoding. A visual overview of this architecture is presented in Fig. 1. Below, we describe each stage in detail and the associated mathematical formulations.

The input image is first processed through the VGG19 network, generating feature maps. The RSPDA method then generates multiple rotated and shifted versions of the feature map, which are concatenated and tokenized into patches. These patches are fed into ViT, which applies self-attention and Feed-Forward Network (FFN) to learn complex spatial relationships. The final representation is passed through a dense layer for classification to predict the skin lesion category.

Data preprocessing and augmentation

All images are rescaled to 168×168 and normalized using the mean/variance calculated from the training set. Normalization is always used for training, validation and test pipelines. We randomly apply stochastic augmentations during training and validation to improve generalization and alleviate overfitting. These augmentations consist of random horizontal flips, small-angle random rotations up to ± 0.02 (about $\pm 7°$) and random zooms of up to 20% in height and width. Figure 2 displays an example of the applied augmentations and demonstrates how one dermoscopic image can appear under different augmentation settings. The test set underwent only resizing and normalization to make the evaluation unbiased.

To prevent data leakage, we ensure that patient-level splitting is followed strictly, i.e., images belonging to the same patient do not occur in more than one subset. This process involves two steps: 80% of the dataset for training, and the remaining 20% is auxiliary data, which is further split between 10% validation and 10% test datasets. The process is randomized but reproducible using a fixed seed (random_state = 101). This approach is designed such that when performance metrics are used to evaluate model generalization, they verify the model's ability to generalize to unseen patients rather than memorizing different views from the same patient.

VGG19 feature extraction

In the first stage of our model, feature extraction is performed using a pre-trained VGG19 network. We selected the VGG19 model for its ability to extract detailed image features, which is critical for medical images where textural properties provide valuable diagnostic information. The input image $x \in \mathbb{R}^{H \times W \times 3}$, where H and W denote the height and width of the input image, respectively, is passed through the VGG19 network.

$$\mathcal{F}(x) = VGG19(x) \tag{1}$$

The $\mathcal{F}(x) \in \mathbb{R}^{H' \times W' \times D}$ represents the extracted feature map, where D is the number of channels. The VGG19 comprises multiple convolutional layers, Rectified Linear Unit (ReLU) activation functions, and max-pooling layers to extract low-level (i.e., edges) and high-level (i.e., patterns) features from the input image. Then, we use the feature output from the last convolutional block $\mathcal{F}(x)$ for further analysis. Figure 3 displays the architecture of the VGG19.

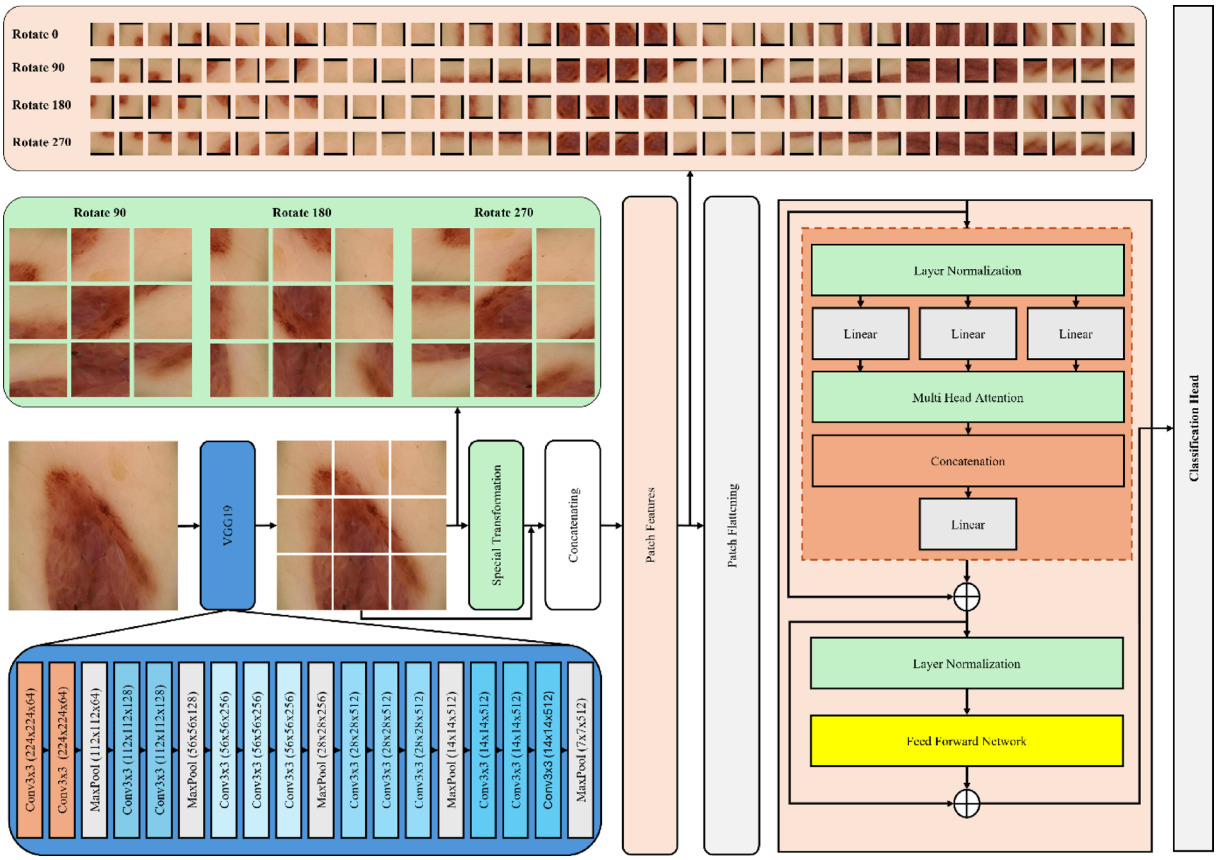


Fig. 1. Overview of the VGG19-RSPDA-ViT architecture.

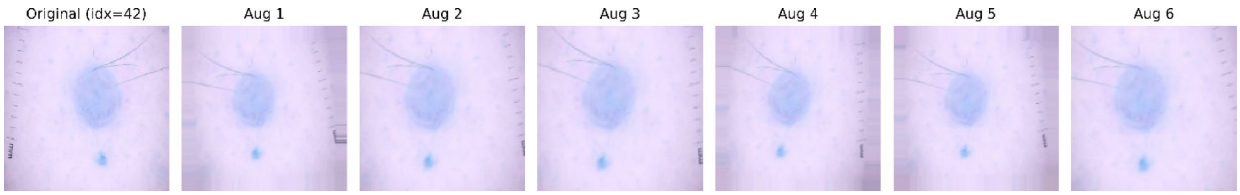


Fig. 2. Illustration of the data augmentation process applied to dermoscopic images.

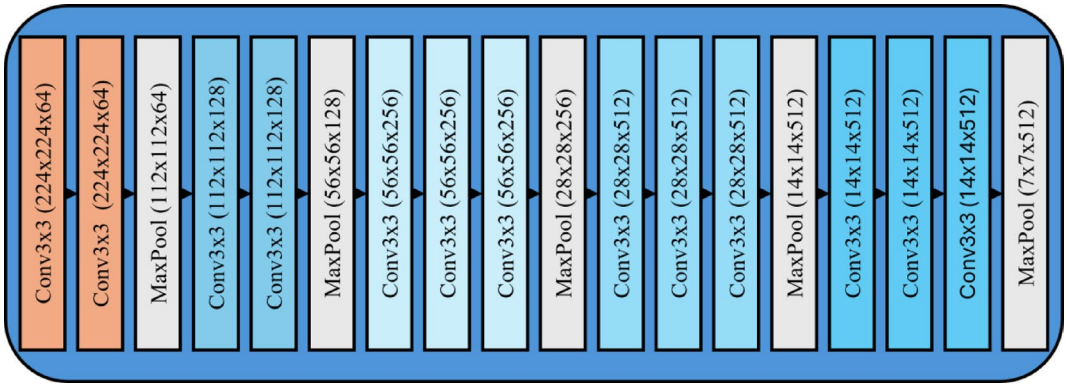


Fig. 3. VGG19 architecture.

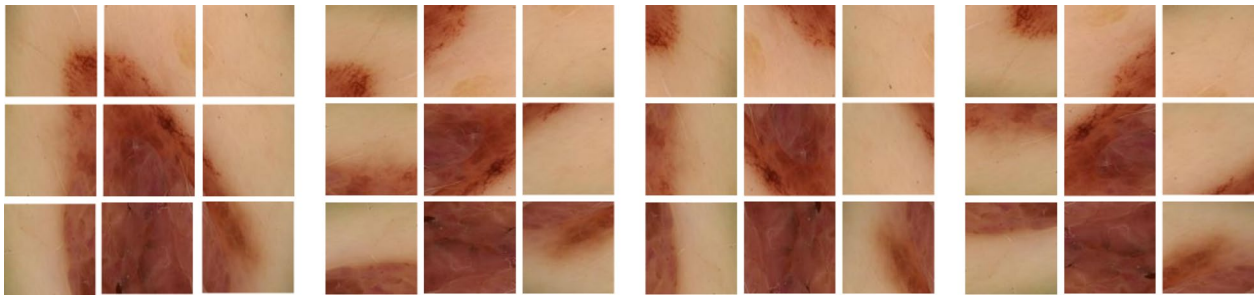


Fig. 4. Rotated samples.

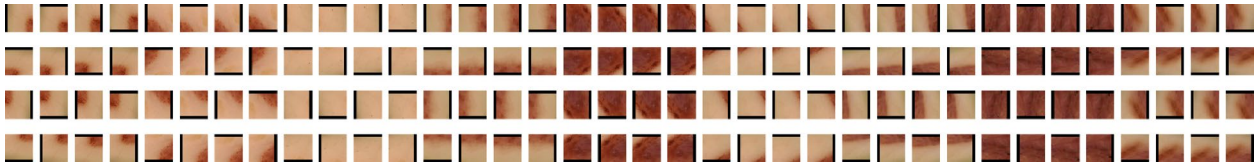


Fig. 5. RSPDA samples.

Rotated and shifted patch data augmentation (RSPDA)

In the second stage, the extracted feature map $\mathcal{F}(x)$ is processed using a Rotated and Shifted Patch Data Augmentation (RSPDA) strategy. This stage introduces rotational invariance and enhances the spatial context for the transformer. The RSPDA method includes the following steps.

Firstly, the feature map $\mathcal{F}(x)$ is rotated by angles $\theta \in (0^{\circ}, 90^{\circ}, 180^{\circ}, 270^{\circ})$, see Fig. 4. Let the feature map $R_\theta(\mathcal{F}(x))$ denote the rotation procedure:

$$R_\theta(\mathcal{F}(x)) = rotate((\mathcal{F}(x)), \theta) \tag{2}$$

For shifting the rotated feature map $R_\theta(\mathcal{F}(x))$, a four-directional shift is employed with d_x and d_y that represent the x-dimension and y-dimension, respectively. We also examine how rotation and translation affect spatial feature uniformity. Equation 3 is used to calculate the $\mathcal{S}_{(d_x, d_y)}$:

$$\mathcal{S}_{(d_x, d_y)}(R_\theta(\mathcal{F}(x))) = shift(R_\theta(\mathcal{F}(x)), d_x, d_y) \tag{3}$$

The rotated and shifted versions of the feature map are then concatenated with the channel dimension to form a single augmented feature map $\mathcal{F}_{(aug)}$, as denoted in Eq. 4:

$$\mathcal{F}_{(aug)} = shift(R_\theta(\mathcal{F}(x)), d_x, d_y) \tag{4}$$

This process generates a feature map $\mathcal{F}_{(aug)} \in \mathbb{R}^{H' \times W' \times D \times N}$, where N is the total number of augmented versions generated through rotation and shift of the feature map. Figure 5 shows the RSPDA augmentation samples, which add valuable complexity to the input data, enabling the ViT to learn spatial relationships better and improve robustness.

The augmented feature map $\mathcal{F}_{(aug)}$ is divided into patches of size $P \times P$. Each patch is flattened and projected into a vector space using the linear layer, as defined in Eq. 5:

$$x_0^i = linear(\mathcal{P}_i) + \mathcal{E}_{pos}^i fori = 1, \ldots, M \tag{5}$$

where $\mathcal{P}_i$ represents the *i*-th patch, $\mathcal{E}_{pos}^i$ is the position embeddings, and M denotes the number of patches. The linear layer maps each patch to a d-dimensional token, and this sequence of tokens is then passed as input to the ViT.

ViT encoding

In the third stage, we feed the sequence of patch tokens through the ViT. The ViT leverages self-attention mechanisms to capture the long-range correlations between patches, allowing the model to capture global spatial information within the skin lesion. Additionally, the self-attention mechanism enables each patch to attend to all the other patches in the sequence, thereby modeling their inter-patch relationships. Let x_0^i denote a given token, the attention score is calculated as Eq. 6:

$$Attention\,(Q,K,V) = SoftMax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{6}$$

where $Q = {}^{i}_{x0}W^{Q}, K = {}^{i}_{x0}W^{K}, V = {}^{i}_{x0}W^{V}$ represent the query, key, and value, respectively, and d_k represents the dimensionality of the keys. The self-attention layer uses multiple heads to capture different aspects of the relationships between tokens using Eq. 7:

$$MHA(_{x0}) = Concat\,(head_1,\ldots,head_h)\,W^{O} \tag{7}$$

where $head_i = \ Attention\,(Q_i,\ K_i,\ V_i)\ \ for\, i \in\ (1,\ \ldots\ ,\ M)$.

After the self-attention layer, the position-wise FFN is applied and calculated by using Eq. 8:

$$FFN(_{x}) = ReLU\,(W_{1x} + b_1)\,W_2 + b_2 \tag{8}$$

where $W_1 and\, W_2$ represent the learned weights and $b_1 and\, b_2$ represents the biases. The output of the FFN is added to the input using the residual connection by Eqs. 9 and 10.

$${}^{i}_{x1} = NormalizationLayer({}^{i}_{x0} + MHA\left({}^{i}_{x0}\right)) \tag{9}$$

$${}^{i}_{x2} = NormalizationLayer({}^{i}_{x1} + FFN\left({}^{i}_{x1}\right)) \tag{10}$$

The process is repeated for $\mathcal{L}$ transformer layers, producing the final representation of each patch token $_{x\mathcal{L}}$. The architecture of the ViT block is displayed in Fig. 6.

Classification head

The output from the transformer is aggregated using Global Average Pooling (GAP), which produces a fixed-size vector representing the entire image. The GAP is calculated by using Eq. 11:

$$_{xfinal} = GAP(_{x\mathcal{L}}) \tag{11}$$

This pooled representation $_{xfinal}$ is then passed through a dense layer with SoftMax activation to produce the class probabilities using Eq. 12:

$$\widehat{y} = SoftMax(W_{_{xfinal}} + b) \tag{12}$$

where $\widehat{y}$ represents the predicted probability distribution over C lesion classes, C represents the number of lesion classes, W represents the weight matrix, and the notation b represents the bias term.

Loss function and optimization

In this study, we used the standard cross-entropy loss without class weighting. Although techniques such as weighted cross-entropy, focal loss, or oversampling could help balance the dataset, we aimed to analyze VGG19-RSPDA-ViT's ability to handle orientation invariance and generalization.

$$\ell = -\sum_{c-1}^{C} {}_{y}c\log\left(\widehat{}_{y}c\right) \tag{13}$$

where $_{y}c$ is the true label (one-hot encoded) and $\widehat{}_{y}c$ is the predicted probability for class C. The Adam optimizer is used for training, with weight decay regularization to prevent overfitting:

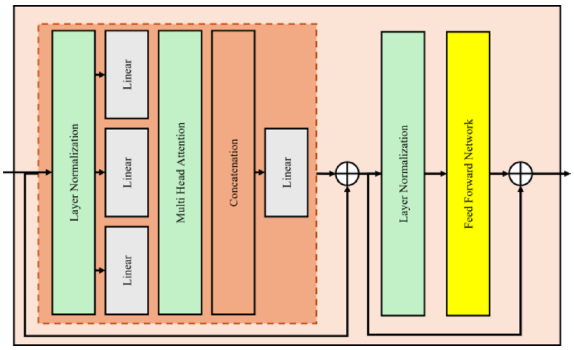


Fig. 6. ViT block.

$$\Theta \leftarrow \Theta - \eta(\nabla_{\Theta}\ell + \lambda \parallel \Theta \parallel_2^2) \tag{14}$$

where Θ represents the model parameters, η is the learning rate, and λ is the weight decay parameter.

Implementation details and hyperparameters

For reproducibility and transparency, this subsection provides a consolidated description of model configuration, optimization settings and training protocol used across all our experiments, see Table 1. Our final architecture combines a VGG19 stem, the proposed RSPDA tokenizer, the ViT encoder and a multi-layer classifier head. A complete overview is provided in Table 1, which consolidates the architectural, augmentation, optimization, and training parameters.

Results and discussion

To evaluate the efficiency and the generalizability of the proposed deep learning model to classify skin lesions, we performed the experiments on three publicly available dermatology datasets widely employed in the literature: HAM10000, MSK10000, and PH2. All collections contain dermoscopic images of skin lesions with a common clinical task. However, they differ in the sources of data collection, the visual content of images, and the population of patients. Our experiments used the standard training, validation and test splits of HAM10000, MSK10000, and PH2 to ensure no data overlap between partitions. Also, 5-fold validation splitting is used for the PH2 dataset. There was no data leakage as duplicate or near-duplicate images were not shared across splits, and data preprocessing steps were added independently within each portion. We also used a full set of threshold-dependent and threshold-independent evaluation metrics to assess model performance. Accuracy, precision, recall, specificity, F1-score, and Receiver Operating Characteristic (ROC) were the main performance metrics.

Training performance analysis

The ability to fit the proposed model on three different datasets sheds light on its generalizability and adaptability across diverse yet domain-related medical data. Figures 7, 8, 9 and 10 offer comprehensive insights into model

Category	Setting	Value / Notes
Input	Image size	$168 \times 168 \times 3$
VGG19 stem	Pretrained	ImageNet (True)
	Trainable	False (frozen during training)
	Tap layer	feature map $21 \times 21 \times 256$ (downsample factor /8)
RSPDA (views)	Rotations	k × 90° with k ∈ [0,1,2,3]
	Shifts (input space)	(dy, dx) ∈ [(0,0), (7,0), (0,7), (7,7)] (via tf.roll, circular shift)
	Views total	16 (4 rotations × 4 shifts)
RSPDA (patching)	Patch kernel/stride (fm)	45,845
	Patch grid per view	$3 \times 3 = 9$ tokens (from 21×21 feature map)
	Tokens total	144 (9 tokens/view ×16 views)
	Token dimension	256 (Conv2D projection + Dense)
	Embeddings	Absolute position (1,144,256); rotation (4,1,256); shift (4,1,256)
ViT encoder	Depth	4 transformer blocks
	Attention	4 heads, key_dim = 64
	MLP per block	256 units (GELU)
	Dropout	0.1
	LayerNorm ϵ	($1e-3$)
Classifier head	Token pooling	GAP
	Dense head	[2048, 1024] with GELU + Dropout 0.3
Optimization	Optimizer	AdamW (TensorFlow ≥ 2.11), fallback Adam
	Learning rate	$1e-4$
	Weight decay	$1e-4$ (AdamW only)
Training	Batch size	16
	Epochs	50
	Callbacks	EarlyStopping (patience 10, monitor = val_auc), ModelCheckpoint (best by val_auc), ReduceLROnPlateau (factor 0.5, patience 3, min_lr $1e-6$)
Loss / Metrics	Loss	Sparse categorical cross-entropy (integer class IDs)
	Metrics	Accuracy, Precision, Recall, Specificity, F1-Score, and ROC
Data split	80:10:10	Patient-level split; no overlap across train/val/test

Table 1. Consolidated hyperparameters and training settings for the proposed VGG19-RSPDA-ViT architecture.

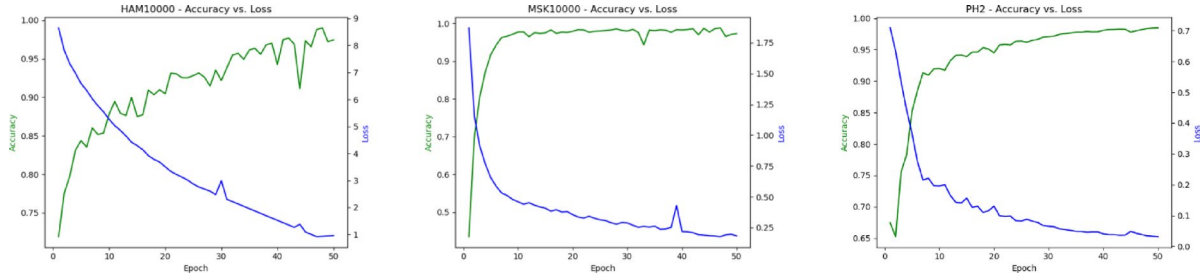


Fig. 7. Validation accuracy and loss per epoch (HAM10000, MSK10000, and PH2).

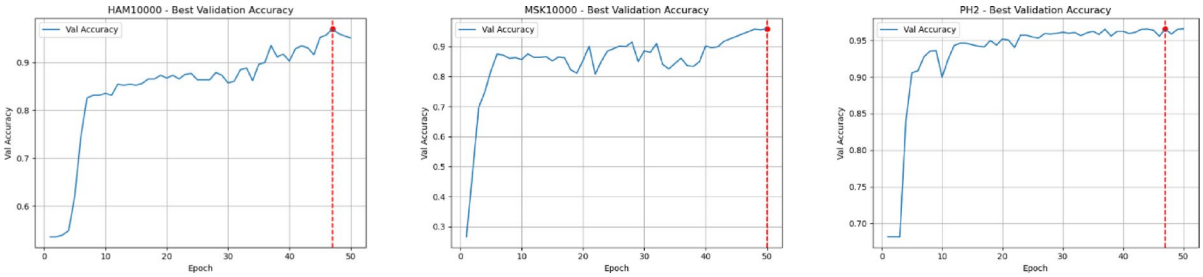


Fig. 8. Best validation accuracy epoch (HAM10000, MSK10000, and PH2).

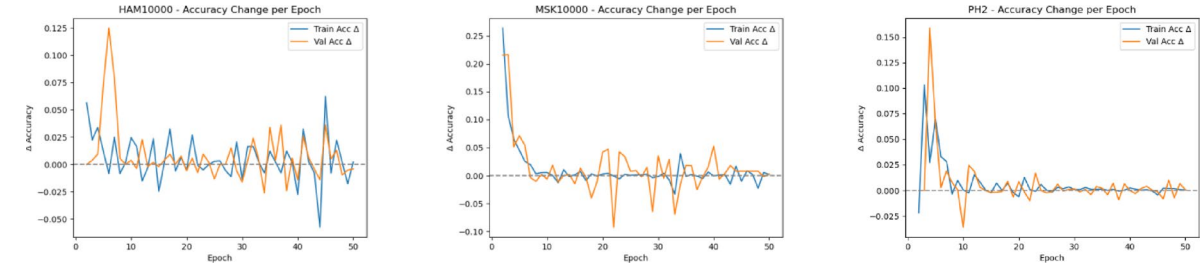


Fig. 9. Delta precision values (HAM10000, MSK10000, and PH2).

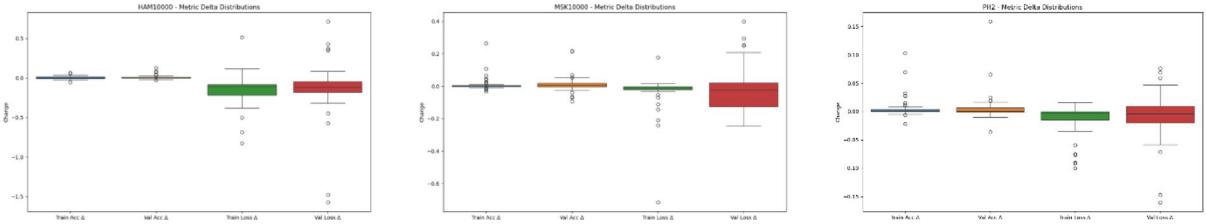


Fig. 10. Accuracy and loss deltas (HAM10000, MSK10000, and PH2).

behavior, stability, and optimization trajectories over training epochs. However, in order to check on the behavior of the model at the training and validation stages, we plotted the validation accuracy and loss curves for each dataset.

As demonstrated in Fig. 7, accuracy gradually improved as loss continuously decreased throughout training. The highest validation accuracy of 0.96 was reached on HAM10000 at epoch 48, while for MSK10000, the highest validation accuracy, 0.95, was obtained at epoch 50. The model showed consistent improvement in the validation accuracy during the early epochs, especially until epoch 10, where the accuracy rose from 0.53 to greater than 0.82.

Subsequently, performance gains became less pronounced, and there were some minor oscillations between 20 and 40 epochs. The highest validation accuracy was around 0.96 at the 48th epoch, as indicated by the red dot

and dashed line, as illustrated in Fig. 8. This result indicates promising generalization with overfitting issues up to the optimal epoch.

Consistent with the HAM10000 observation, the performance gain of the model on the MSK10000 dataset increases significantly in the first few epochs and achieves about 0.84 at the fifth epoch. We observed continued gains in the subsequent training epochs: the highest validation accuracy of about 0.95 was obtained on the 50th epoch. This suggests that our model benefited from a comprehensive training schedule and converged to a powerful solution. For generalization, we measured the differences between training and validation at the best epoch of each dataset. As shown in Fig. 8, the small delta precision values between datasets suggest good generalization with minimal overfitting.

Figure 9 shows the delta distributions (epoch-to-epoch changes) of accuracy and loss for both datasets. In MSK10000, the training and validation metrics are relatively concentrated and stable. The consistency between epochs indicates a stable optimization trajectory and strong convergence. On the other hand, HAM10000 shows larger interquartile ranges and a higher frequency of outliers, particularly in validation loss deltas, indicating less stable generalization. Furthermore, to investigate the degree of overlap between loss and accuracy in the training process, we also use a dual-axis plot in Fig. 10. This indicates that validation loss and accuracy are significantly negatively correlated across both datasets, confirming that validation accuracy is a reliable convergence indicator.

Figure 10 illustrates the accuracy and loss deltas of HAM10000 and MSK10000 datasets, which measure the differences in model performance at the best epoch for each run. For both datasets, the accuracy deltas are close to zero for training, and the validation datasets show that the learning process is stable and the generalization is strong. The loss differences show slightly more variation, but within a reasonable range. For HAM10000 results, the deltas of validation loss exhibit a greater spread, and lastly, there are a few outliers, indicating a small amount of variation in the convergence of the optimization. These fluctuations did not result in large performance variations, as the accuracy delta was consistent. In contrast, we observe that the delta distributions are relatively narrow for all metrics, including validation accuracy for the MSK10000 dataset, which suggests that the training procedure is highly congruent and reproducible. These experiments support the significance of the model and reliable generalization on both datasets.

Confusion matrix analysis

Furthermore, to evaluate class-wise performance, confusion matrices were generated for each dataset. Figure 11 shows the confusion matrices of the three datasets to demonstrate the classification performance of the two models.

Our approach performed well for MSK10000, successfully differentiating between benign and malignant cases with good class separation and had only a few misclassification cases. This demonstrates good binary classification performance and the stability of the decision boundaries. For the HAM10000 dataset with a more complex multi-class classification problem, the confusion matrix shows a good accuracy for the majority of the classes. The model performed well in the detection of frequent classes of lesions, with some confusion between some classes due to clinical similarities. This fine-grained performance is consistent with clinical diagnostic challenges and indicates that the model can capture subtle variations in the dermatoscopic data. However, the model still struggles with certain classes that are difficult to distinguish from one another.

Detailed class performance metrics for both datasets are shown in Table 2. The model achieved consistent balanced results for benign and malignant categories on MSK10000 with an overall accuracy, precision, recall, specificity and F1-score of 97.9%. The model performed best for malignant types MEL (sensitivity 97.0%, specificity 99.7%), BCC (sensitivity 96.2%, specificity 99.8%) and AKIEC (sensitivity 100%), all of which had an AUC greater than 0.9 on HAM10000. The specificity was above 98% for all classes, demonstrating consistency in distinguishing between benign and malignant lesions. For underrepresented cases such as VASC, a lower sensitivity (60%) was observed due to dataset imbalance, and still, high precision and specificity for this class were attained (92.3% and 99.2%).

Classification-specific metrics also provide more clinical interpretability. High sensitivity for malignant lesions is important to avoid missed diagnoses, and high specificity ensures reliable identification of benign lesions. The low sensitivity of VASC lesions, given their rarity, emphasizes the challenge of handling data

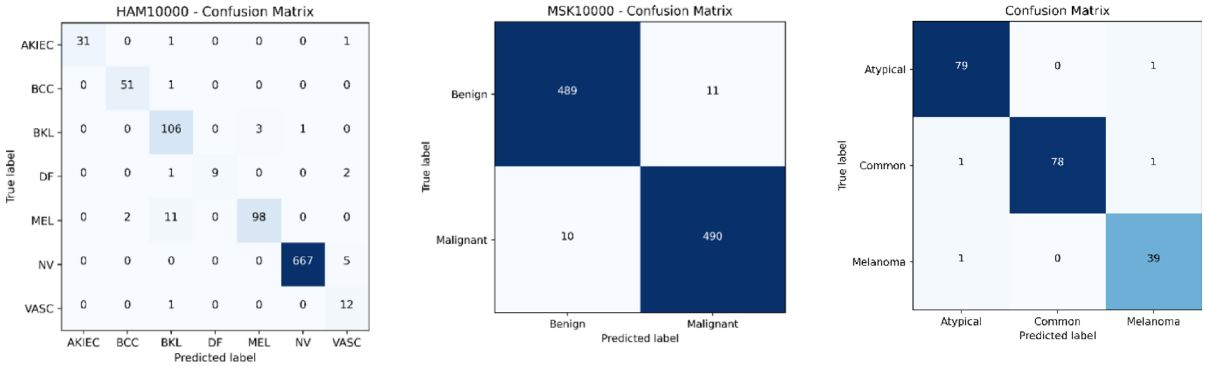


Fig. 11. Confusion matrix (HAM10000, MSK10000, and PH2).

Dataset	Class	TP	TN	FP	FN	Accuracy	Precision	Recall	Specificity	F1 Score
MSK10000	Benign	489	490	11	10	97.90	97.80	98.00	98.00	97.90
	Malignent	490	489	10	11	97.90	98.00	97.80	97.80	97.90
HAM10000	AKIEC	31	970	2	0	99.80	93.94	100	100	96.88
	BCC	51	948	1	2	99.70	98.08	96.23	99.79	97.14
	DKL	106	875	4	15	98.10	96.36	87.60	98.31	91.77
	DF	9	972	3	0	99.70	75.00	100	100	85.71
	MEL	98	884	13	3	98.40	88.29	97.03	99.66	92.45
	NV	667	326	5	1	99.40	99.26	99.85	99.69	99.55
	VASC	12	982	1	8	99.10	92.31	60.00	99.19	72.73
PH2	Atypical	79	2	1	118	98.50	97.53	98.75	98.33	98.14
	Common	78	0	2	120	99.00	100	97.50	100	98.73
	Melanoma	39	2	1	158	98.50	95.12	97.50	98.75	96.30

Table 2. Summarization of per-class performance metrics for both HAM10000, MSK10000, and PH2 datasets.

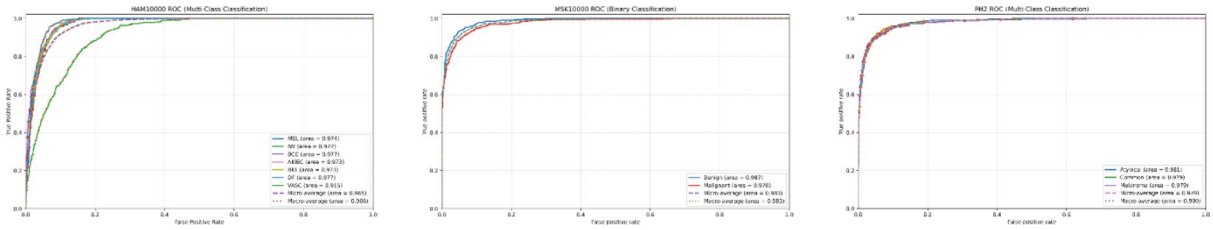


Fig. 12. ROC curves (HAM10000, MSK10000, and PH2).

imbalance in this domain, which could be tackled by a more balanced dataset, integrating other modalities or increasing annotation size.

Binary and multi-class classification analysis

ROC curves and relative AUC values were also calculated to support these findings. For MSK10000, AUCs reached 0.987 (Benign) and 0.978 (Malignant), respectively, with macro- and micro-averages of 0.983. For HAM10000, per-class AUCs were in the range 0.915 (VASC) to 0.977 (BCC, DF) with macro-average = 0.966 and micro-average = 0.965.

From Fig. 12, the model achieved an AUC of 0.983 for both the macro and micro average AUC, with the benign class having values of 0.987 and the malignant class having values of 0.978in terms of a binary classification task (benign versus malignant) in the MSK10000 dataset. Our results support that the model we created may be useful for clinicians in making binary diagnostic decisions about whether a lesion is or is not a melanoma (high sensitivity combined with high specificity).

The per-class AUCs for HAM10000 indicate the robustness of our model for malignant class separation compared to other supervised approaches. This is expected as the underrepresented VASC class has the lowest AUC (0.915), which correlates with its substantially lower sensitivity in the per-class performance table, Table 2, which reflects a common challenge in real-world dermatological datasets due to class imbalance.

Model explanability analysis

To further validate the robustness of our predictions, we visualize the interpretability of the VGG19–RSPDA–ViT framework through Grad-CAM. This technique produces visual explanations by marking the image areas that contribute the most to the model's predictions. This is an essential visual confirmation technique for clinical AI applications; it helps clinicians confirm whether the automated system focuses on lesion areas or background artifacts that are irrelevant. Figure 13 displays an example of the Grad-CAM visualization.

To improve clinician trust and interpretability, we included explainability analysis based on Grad-CAM and Transformer attention maps. The visualizations in Fig. 13 verify that the model focuses attention on the lesion region, which correlates with features dermatologists might tend to look at. Crucially, any irrelevant background structures (e.g., skin or hair) were largely disregarded, indicating that the model was making predictions based on clinically significant features. These findings enhance the interpretability of our model and also show the potential of integration of the model with a clinical workflow.

Although the Grad-CAM visualizations generally captured the regions of the lesion, in certain images, the heatmaps seemed to be more scattered, also reaching up to the background artifacts (e.g., hair), in line with a known limitation of the current saliency-based methods. Future work will address this limitation by examining other interpretable methods, such as the Transformer attention maps or the SHAP, for learning more interpretable features.

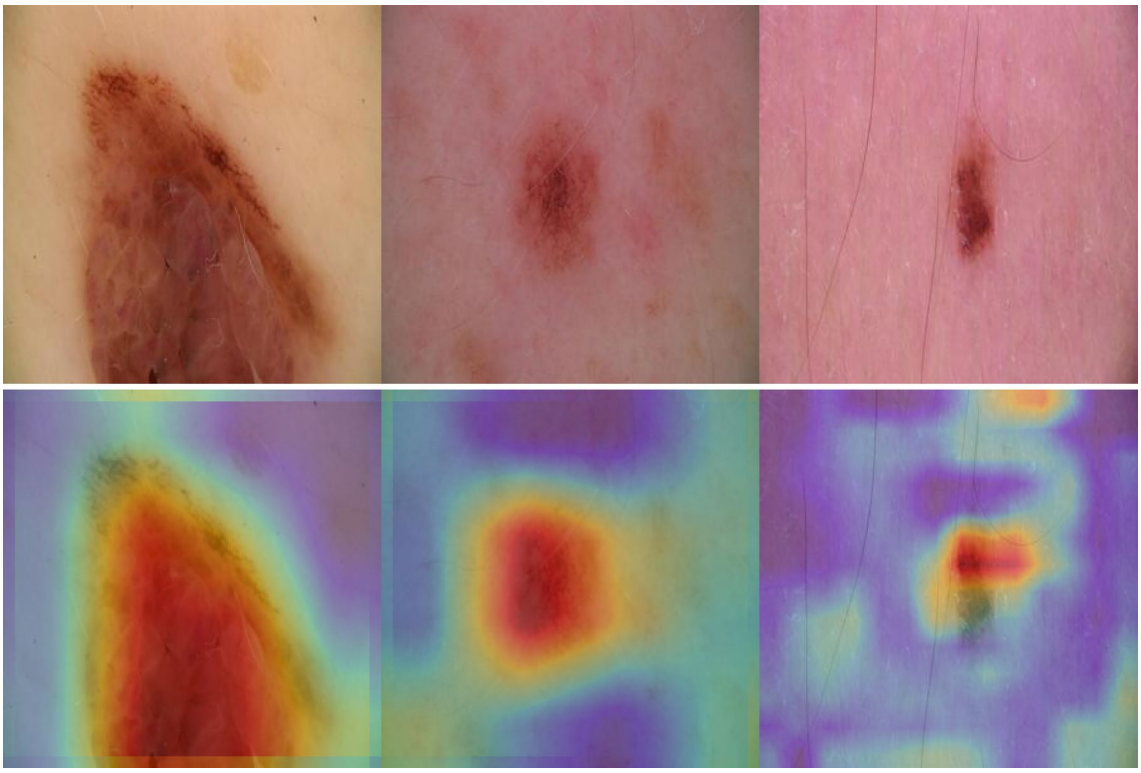


Fig. 13. Example Grad-CAM visualizations.

Authors. (year) Ref.	Datasets	Accuracy	Precision	Recall	Specificity	F1 score
Sinchana et al. (2023)[36]	MSK10000	94.57	87.00	89.00	87.00	-
Bhuvaneswari and Raman (2023)[39]	MSK10000	91.10	-	90.50	-	-
Waheed et al. (2023)[7]	MSK10000	91.00	-	-	-	-
Adebiyi et al. (2024)[33]	HAM10000	94.11	90.73	90.19	98.33	90.33
Ghosh et al. (2024)[32]	MSK10000	91.60	91.60	91.60	-	91.60
Ingle and Shaikh (2024)[34]	HAM10000	88.83	89.00	89.00	-	89.00
Sabir and Mehmood (2024)[37]	MSK10000	97.00	95.00	93.00	99.00	-
Sadi Al Huda et al. (2025)[38]	MSK10000	93.40	-	-	-	-
Srivastava et al. (2025)[40]	MSK10000	93.86	93.97	93.86	96.72	93.85
Aruk et al. (2024)[41]	HAM10000	94.30	93.00	89.59	-	91.11
Yang et al. (2023)[42]	HAM10000	94.10	-	-	-	-
Ghazouani (2025)	PH2	97.00	-	-	-	97.00
Golkarieh et al. (2025)	PH2	89.53	-	90.97	-	90.62
VGG19-RSPDA-ViT (ours)	MSK10000	97.90	97.90	97.90	97.90	97.90
	HAM10000	97.11	91.52	91.89	99.52	90.89
	PH2	98.67	97.55	97.92	99.03	97.72

Table 3. Summarization of the overall performance metrics for both HAM10000 and MSK10000 Datasets.

Comparative analysis

Finally, the performance of the proposed VGG19-RSPDA-ViT model is compared with the state-of-the-art methods on the MSK10000, HAM10000, and PH2 datasets, as presented in Table 3. Although Table 3 summarizes the results from MSK10000, HAM10000, and PH2, one should be aware that variations in dataset partitions, preprocessing pipelines, and evaluation protocols across the studies may hinder comparability. Therefore, our objective is not to claim superiority over all prior works, but to propose VGG19-RSPDA-ViT as a model that can address the two benchmark datasets with consistently good performance under a single experimental setup.

For the MSK10000 dataset, previous works have provided a mixed performance across different metrics, suggesting different strengths in precision, recall, and specificity. Although some models performed well according to individual metrics, few achieved consistently high performance across all key evaluation criteria.

In comparison, our proposed method demonstrates good binary classification capability in terms of accuracy, precision, recall, specificity and F1 score.

Generally, previous works on HAM10000 have achieved variable performance due to the more complex multi-class classification problem. Although some models achieved good precision and recall results, consistent performance across all metrics was rarely observed. In contrast, the proposed approach shows comparable performance with better results in specificity, precision, and recall. This outcome demonstrates its capability to differentiate multiple types of skin lesions and has the potential for clinical use in various dermatologic scenarios.

Also, the proposed VGG19-RSPDA-ViT model reported on 98.67% accuracy and a F1-score of 97.72%, with all of the precision, recall and specificity crossing the range higher than 97% on the PH2 image dataset. This result is better than the results obtained by Ghazouani, where accuracy and F1-score of 97.00%, but it also overcomes Golkarieh et al. with 89.53% accuracy, 90.97% recall and 90.62% F1-score. For the state-of-the-art former, a + 1.67% accuracy improvement with respect to Ghazouani is an indicator of the success for designing a hybrid scheme; while a significant gain by more than 9% in accuracy and greater than 7% in F1-score with Golkarieh et al. and its strength in dealing with the variation of lesions in the PH2 dataset. Furthermore, the balanced distribution of precision, recall, and specificity indicates that the proposed model does not favor one class over another, thereby ensuring consistent generalization across lesion categories.

Recent TransMed, MedT and Swin-UNet variants of CNN-ViT hybrids have shown great potential in medical image analysis. However, many of these architectures are specifically designed for segmentation or are computationally intensive multimodal fusion pipelines, computational intensive processing modules that might hinder interpretability and clinical utility. Our approach is distinct in three major ways: (i) it utilizes the lightweight, yet easily-interpretable VGG19 backbone for dermoscopic feature extraction; (ii) augments data through a novel RSPDA scheme that enhances rotational invariance and robustness beyond standard augmentation; and (iii) exhibits task transferability between clinical scenarios, as shown by superior performance in both binary and multi-class skin lesion classification tasks. These differences highlight the technical novelty of the work we performed, in particular for clinical decision support systems where simplicity, robustness and reproducibility are crucial. ROC analysis on both datasets was conducted further to assess the discriminative power of the proposed VGG19-RSPDA-ViT. ROC curves for MSK10000 and HAM10000 are shown in Fig. 12. The ROC curve is used to plot the True Positive Rate (Recall) against the False Positive Rate (1 - Specificity) over all decision thresholds, which gives a threshold-independent evaluation of classifier performance.

In spite of the good performance of the proposed VGG19-RSPDA-ViT model, there are some failure cases reported in comparison to the multi-class HAM10000 dataset. Many misclassifications were between clinically similar categories like melanoma and nevi or AKIEC with BCC, which mirrored the visual overlap in dermoscopic patterns that challenge even dermatologists.

Cross-validation classification

To test the generalization performance of the proposed model and avoid overfitting to biased splits, we used a 5-fold cross-validation method in PH2. The data was partitioned into five different sets at random. Four of the folds were used as training data on each iteration, and one fold was left for validation. This process was repeated five times, so that each time a sample was used to validate one and the other four times for training. Mean and Standard Deviation (SD) across the folds are reported as the last result metrics.

The Validation curve shows that with repetitions, the model converged on average as expected over the folds. Validation accuracy, Fig. 14, increased quickly starting from the first 10 epochs, beyond 90%, before leveling up to at least converging to a range: 95–98%.

Validation loss, Fig. 15, exhibited the same tendency, with all of them being consolidated around 95%, whereas Validation F1-score, precision, and recall, Fig. 16a–f, showed equally good F1-score, precision, and recall levels.

Validation-precision remains high throughout training, reaching values of more than 0.90, demonstrating that the model can lower false positives. Validation recall is more variable in the beginning epochs; however, it also converges to more than 0.90, indicating consistent sensitivity and less probable false negatives. The verification F1-score also follows the tendency of precision and recall, reaching greater than 0.90, and becomes stable, which

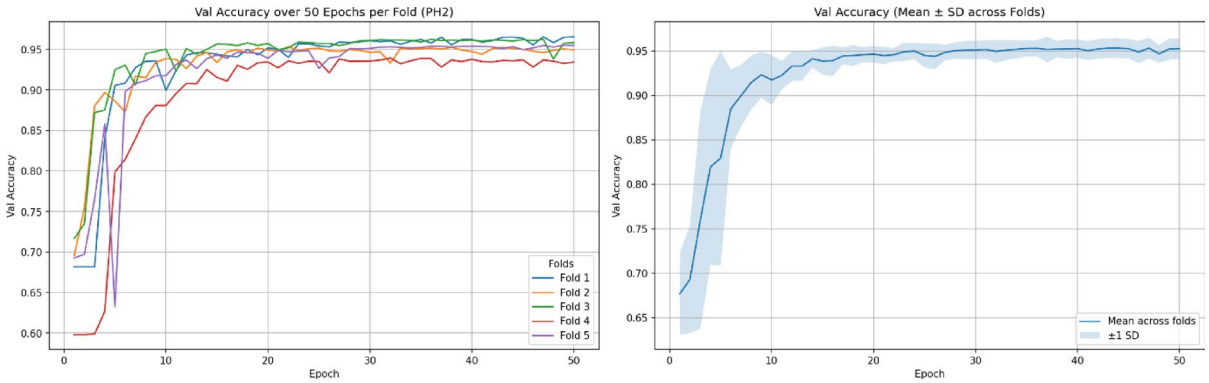


Fig. 14. Validation accuracy over 50 epochs across the five folds (left) and the Mean ± SD across folds (right).

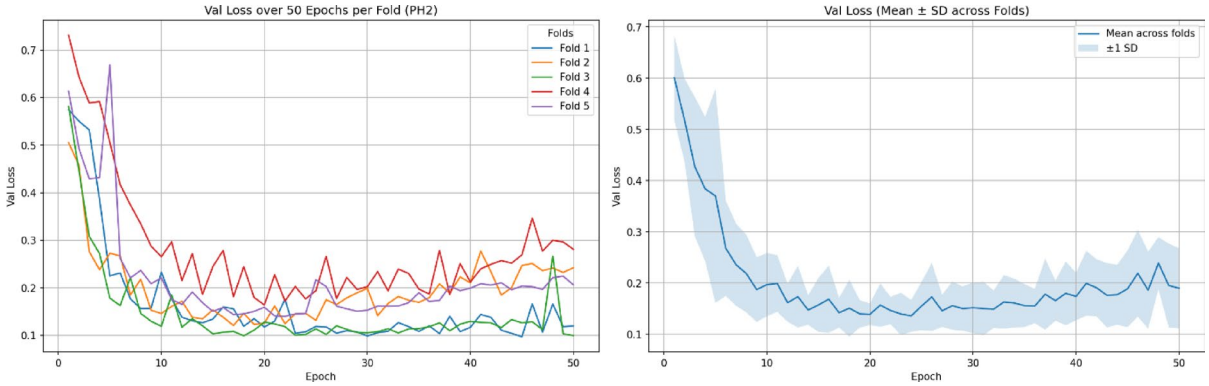


Fig. 15. Validation loss over 50 epochs across the five folds (left) and the Mean ± SD across folds (right).

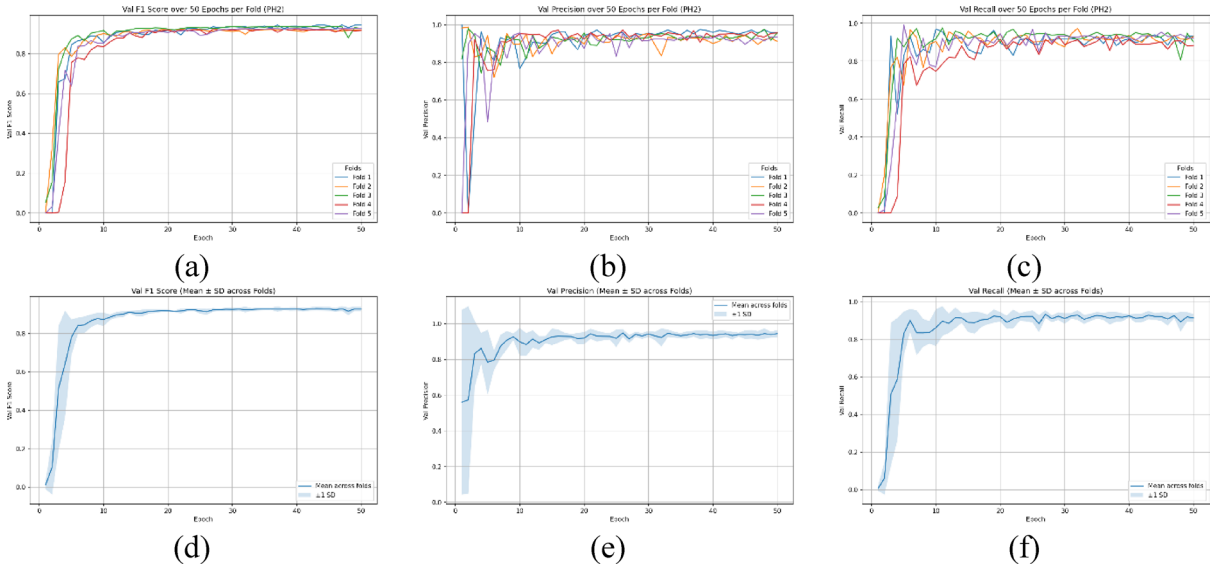


Fig. 16. (**a**–**c**) Validation F1-score, precision, and recall over 50 epochs across the five folds and (**d**–**f**) the Mean ± SD across folds.

Metric	Training (Mean ± SD)	Validation (Mean ± SD)
Accuracy	97.5 ± 0.8	96.2 ± 0.9
Precision	97.8 ± 0.7	96.0 ± 1.0
Recall	97.3 ± 0.9	95.8 ± 1.1
F1-score	97.6 ± 0.8	96.0 ± 0.9
Loss	0.05 ± 0.01	0.12 ± 0.02

Table 4. Cross-validation performance of the proposed model on the PH2 dataset.

indicates the model's sensitivity matched with specificity. Furthermore, the cross-validation performance of the proposed VGG19-RSPDA-ViT on the PH2 dataset is illustrated in Table 4.

The Mean ± SD plot also shows that the variability between folds was characteristic in the early stages of training, but significantly diminished upon convergence. This decrease in variance emphasizes that the response is robust and stable after having cross-validation for different splits. Taken together, these findings demonstrate that the model generalizes well, exhibiting high and stable validation scores across all three primary. Table 5 shows the performance of our proposed model using the 5-fold cross-validation. Training metrics were slightly higher. However, these small differences between training and validation scores, together with low standard deviations, suggest that the model generalizes well across folds with no sign of overfitting. The validation loss converged to 0.12 ± 0.02, while the training loss remained around 0.05 ± 0.01, indicating stability of convergence

Model	Tokens	Flattened features	Parameters (Total)	Trainable params	Efficiency	Training time per epoch
VGG19-ViT	9	2,304	10,732,354 M	8.4 M	Compact, efficient	845s
VGG19-RSPDA-ViT	144	36,864	81,547,330 M	79.2 M	Heavy, robust	3835s

Table 5. Model complexity comparison.

for all runs. This confirmed the model to hold high stability of classification performance irrelevant of different PH2 dataset subsets evaluation metrics.

Limitations and future work

A limitation is that we have not performed cross-validation or validated our results with other independent clinical cohorts. While our model provided strong performance on two benchmark datasets, indicating its generalizability, validation in larger independent clinical cohorts or via using k-fold cross-validation would provide more evidence of its generalizability. We see further exploration in this direction as important future work. Finally, it is crucial to mention that the nature of the HAM10000 dataset is imbalanced by default, and certain types of lesions are underrepresented. Most classes maintained a precision-recall balance with our approach, although minority categories were sometimes misclassified; this is due to the difficulty of distinguishing between rare lesion types. Although we did not use explicit balancing methods (e.g., weighted loss or focal loss) in our framework, applying them might help to improve classification performance for imbalanced classes in future work.

The VGG19-RSPDA-ViT model may be further developed for incorporation into clinical pipelines as an assistive decision-making system. This could also assist in daily dermoscopic workflows for dermoscopic diagnosis by reducing inter-observer differences, providing a rapid and reproducible second opinion of the dermoscopic interpretation. In teledermatology scenarios, automated triage of lesions could enable pre-filtering of high-risk cases for priority expert review, thus facilitating quick diagnosis in underserved areas. Outside of specialist use, the system may be mounted as a pre-screening module at the primary or community care level, where it could aid non-expert clinicians in identifying lesions for referral. Several additional steps are needed for safe clinical adoption, including prospective validation in a range of clinical cohorts, regulatory approval, and the creation of user-friendly interfaces that can be easily added to the existing electronic health record and imaging systems.

Ethical considerations are essential in the development and implementation of AI-based skin cancer diagnostic tools. While HAM10000 MSK10000, and PH2 are public benchmark datasets in which patient data has been de-identified, real-world medical applications must adhere to strict privacy and consent regulations (e.g., HIPAA, GDPR). In addition, all datasets present biases towards lighter skin tones and specific acquisition devices, which could exacerbate algorithmic bias against underrepresented populations. Misdiagnosis also has strong ethical implications, as false negatives could delay identification of malignant lesions, while false positives could cause unnecessary patient anxiety and treatment. We therefore emphasise that the proposed system should serve as a decision-support tool to dermatologists, not as a replacement for them. Prospective validation in diverse cohorts with various skin tones and careful clinician oversight of any such system is essential to ensure the system is equitable, transparent, and accountable.

Ablation study

To investigate the contribution of the proposed RSPDA, we compare two variations of our framework: (i) VGG19-ViT and (ii) VGG19-RSPDA-ViT. This comparison allows us to examine RSPDA's influence on feature representation and classification performance.

Model structure comparison

The VGG19-ViT model uses a VGG19 backbone and applies a basic patchification operation that extracts nine tokens from a 3×3 grid across the feature map. These tokens are processed through 4 Transformer encoder blocks, linearized, and fed through a two-layer MLP with 2048 and 1024 hidden units for classification. This creates a compact structure with about 10.7 million parameters, of which 8.4 million are trainable. Most parameters are concentrated in the classifier layers. The model is lightweight and efficient since the Transformer blocks process only nine tokens. However, using only nine tokens limits spatial diversity, and the absence of rotation or shift augmentations prevents learning orientation-invariant features.

In contrast, the VGG19-RSPDA-ViT variant replaces simple patchification with a tokenizer that performs four rotations and four pixel shifts, providing 16 different views per image. Each view generates 9 tokens, resulting in 144 tokens total. They are then passed into the same four-layer Transformer encoder. However, the quadratic complexity of self-attention significantly increases computational cost: while the baseline requires only 81 pairwise interactions per layer, RSPDA requires over 20,000. The size of the flattened feature representation also grows significantly, from 2304 dimensions in the baseline to 36,864 in the RSPDA model. When flattened into the 2048-unit dense layer, this yields a single weight matrix with over 75 million parameters, bringing the total parameter count to 81.5 million, nearly eight times larger than VGG19-ViT. Consequently, VGG19-RSPDA-ViT has significantly higher memory and computational requirements than VGG19-ViT for both training and inference. Table 5 presents the model complexity comparison between the VGG19-ViT and the VGG19-RSPDA-ViT.

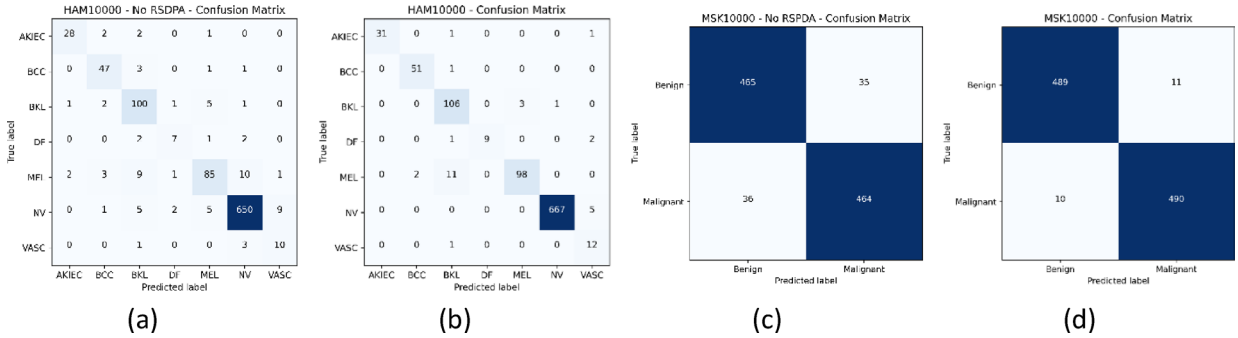


Fig. 17. Confusion matrix analysis (**a**) and (**c**) VGG19-ViT (HAM10000 and MSK10000) and (**a**) and (**c**) VGG19-RSPDA-ViT (HAM10000 and MSK10000).

Model	Dataset	Accuracy	Precision	Recall	Specificity	F1 score
VGG19-ViT	MSK10000	92.90	92.90	92.90	92.90	92.89
VGG19-RSPDA-ViT	MSK10000	97.91	97.90	97.90	97.90	97.90
VGG19-ViT	HAM10000	92.33	79.36	81.31	98.34	79.95
VGG19-RSPDA-ViT	HAM10000	97.10	91.52	91.89	99.52	90.89

Table 6. Performance comparison of the VGG19-ViT model and our proposed VGG19-RSPDA-ViT model on MSK10000 and HAM10000.

Confusion matrix analysis

Figure 17a,c show the confusion matrices of the VGG19-ViT model on HAM10000 and MSK10000, while the corresponding confusion matrices for our proposed VGG19-RSPDA-ViT model are shown in Fig. 17b,d.

On MSK10000, the baseline model misclassified 35 benign lesions as malignant and 36 malignant lesions as benign. In contrast, the RSPDA model made significantly fewer errors, maintaining a better balance between sensitivity and specificity. For HAM10000, the baseline model showed particular confusion between melanoma (MEL) and benign nevi (NV), as well as among rarer classes like AKIEC and BCC. RSPDA substantially reduced inter-class misclassifications, improving recall for minority classes.

VGG19-RSPDA-ViT outperformed VGG19-ViT in accuracy, recall, and F1-score despite its computational overhead. The network learns robust orientation-invariant representations through rotated and shifted patches while explicitly preserving enhanced spatial context. However, these benefits come at a significant computational cost.

Comparative results

To evaluate RSPDA's specific contribution, we compared the baseline VGG19-ViT model with our proposed VGG19-RSPDA-ViT. As shown in Table 6, RSPDA improves accuracy, precision, recall, specificity, and F1-score on both MSK10000 and HAM10000 datasets. This demonstrates that RSPDA's robustness to orientation and translation variations significantly contributes to the model's superior performance.

The results confirm the trade-off between efficiency and performance. The VGG19-ViT model is computationally lighter and faster but lacks sensitivity and suffers from inter-class confusion. VGG19-RSPDA-ViT achieves substantial improvements in accuracy, recall, and robustness across both datasets.

Conclusions

In this study, we established a new hybrid VGG19-RSPDA-ViT model for automatic skin lesion classification that leverages the fine-grained local feature extraction capabilities of VGG19 and the global contextual learning capacity of ViT, enhanced by our proposed RSPDA scheme. Notably, unlike handcrafted image-level augmentations, RSPDA systematically generates rotated and shifted feature-map views before tokenization, enriching the sequence input to the transformer and enabling orientation-invariant representation learning. Through extensive experiments on the HAM10000 and MSK10000 benchmark datasets, our model achieved impressive performance with 97.1% and 97.9% accuracy, respectively. Ablation studies demonstrate that all three components (VGG19 backbone, ViT integration and RSPDA augmentation) contribute significantly, with RSPDA providing the most significant improvement.

Our comprehensive evaluation included ROC curves, class-wise sensitivity/specificity, and interpretability analyses. Grad-CAM visualizations demonstrate that the model consistently focuses on clinically relevant lesion regions and suppresses irrelevant background structures. These results highlight both the predictive performance and interpretability of our approach, which are essential for clinical adoption.

However, several limitations should be acknowledged. First, while HAM10000 and MSK10000 differ in acquisition conditions and demographics, our model shows limited generalization to rare lesion classes. Second,

larger, more diverse clinical validation cohorts, with broader skin tone representation and varied imaging conditions, are necessary to validate generalizability fully. Additionally, we did not perform user studies, domain expert evaluations, or pilot deployments, which are crucial for assessing usability and integration into real-world clinical workflows. Finally, although inference speed on GPU is near real-time, further optimization would be needed for resource-constrained scenarios (e.g., mobile dermoscopic).

Future directions

Based on this work, a few directions for future research would be worth pursuing. First, we aim to validate in additional and diverse cohorts (Derm7pt, ISIC challenge datasets, as well as clinical in-house collections), further to substantiate the robustness and generalization of our approach. Second, we seek to investigate semi-supervised and self-supervised learning approaches for more effectively using large unlabeled dermoscopic images. Third, using multimodal information, such as patient metadata, clinical images, and histopathology, for Desktop Routine may further improve diagnostic performance. Last but not least, optimization of the framework for real-time clinical application and performing a reader study with dermatologists in a prospective way are important intermediate steps toward clinical translation.

Data availability

The data supporting the findings of this study are completely available online. The Melanoma Skin Cancer Dataset of 10000 Images (MSK10000) dataset: [https://www.kaggle.com/datasets/hasnainjaved/melanoma-skin-cancer-dataset-of-10000-images] (https://www.kaggle.com/datasets/hasnainjaved/melanoma-skin-cancer-dataset-of-10000-images), the Human Against Machine with 10000 training images (HAM10000) dataset: [https://dataverse.harvard.edu/dataset.xhtml? persistentId=doi:10.7910/DVN/DBW86T] (https://dataverse.harvard.edu/dataset.xhtml? persistentId=doi:10.7910/DVN/DBW86T), and the PH2 dataset: https://www.fc.up.pt/addi/ph2%20database.html.

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Author contributions

F. H. N. participated in the concept and design, model structure design, data preparation, and writing of the original draft. Z. N. K. helped with the literature review, manuscript structuring, and validation results. F. M. project oversight, overall study design, review and editing of the manuscript, model generation, data curation, and formal analysis. M. S. M. R. contributed to optimizing deep learning, evaluating the models, data analysis, visualization and resources. V. S. C. final manuscript review, proofreading, and critical revision. A. H. A gave technical support and helped implement the proposed method and evaluation metrics.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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